B. Tech. (Computer Engineering) Vth Semester

2nd Sessional 2022-23, AUTOMATA THEORY (CEN-502)

Time: 1 Hour **MM:15**

Note: Attempt all questions. **Date: 01/11/23**

Your Roll No : .....

Q No.	CO Mapped
1	3
2	3
3	4

Q. No	Question	Marks
1.	Construct the CFG for the following language with suitable verifications : a. L1 = { w | #0 (in w) != #1 (in w), and w ∈{0, 1}*} b. L2 = { w | #a (in w) is odd, #b (in w) is even, and w ∈{a, b}*} ***You Must Verify your answer with suitable examples.***	4
2.	Convert the following CFG to CNF along with mentioning appropriate rules wherever needed. S → aAbB | AC b A → AbB | ε B → CAC | ε C → d | ε	5
3	Explain the process of converting a CFG to PDA along with its merits and demerits. And also convert the resultant CNF of Q2) to its equivalent PDA using this process. Finally verify whether the string "***abbd***" is acceptable by the resultant PDA or not using the Tabular approach showing the contents of stack and input at each step.	6